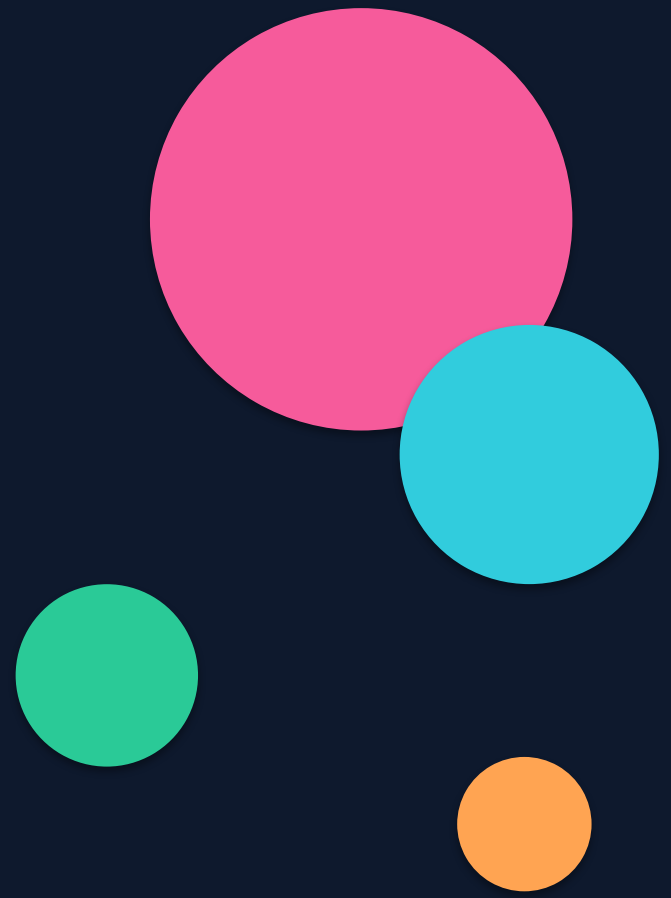
Advanced Data Science & Applications

Detailed compact notes with concepts, methods, examples and practical reminders.



6 topic slides + cover = 7 slides

Ensemble Learning and Feature Methods

Key concepts • practical context • quick revision

- Decision trees split data using feature rules and are often easy to explain.
- Random forests average many trees trained with randomness to reduce variance.
- Boosting builds models sequentially, focusing later models on earlier errors.
- Feature selection removes less useful inputs; dimensionality reduction compresses information.
- Ensembles may improve accuracy but can increase compute and reduce interpretability.
- Evaluate methods on unseen data and tune complexity to avoid overfitting.

EXAMPLE / REMEMBER

Choose the trade-off among performance, speed, interpretability and maintenance.

Deep Learning Basics

Key concepts • practical context • quick revision

- Neural networks combine layers of weighted operations with activation functions.
- Training adjusts weights to reduce a loss function, commonly using gradient-based optimization.
- Convolutional neural networks are widely used for image and spatial data.
- Sequence models and attention-based architectures support text, time series and other ordered data.
- Deep learning often needs substantial data, compute and careful validation.
- Transfer learning reuses pretrained representations when they suit the target task.

EXAMPLE / REMEMBER

A neural network is not automatically better than a simpler model on small tabular data.

Time Series, Big Data and Modern Workflows

Key concepts • practical context • quick revision

- Time series contain ordered observations; trend, seasonality and changing variance matter.
- Forecasting must respect time order when splitting training and test periods.
- Big-data systems distribute storage and computation across machines when one machine is insufficient.
- Batch processing handles accumulated data; streaming systems process events continuously.
- Cloud platforms provide scalable storage, compute, managed databases and model-serving options.
- Reproducible workflows version code, data definitions, features, parameters and model artifacts.

EXAMPLE / REMEMBER

Avoid random train/test splits for forecasting when they expose future data to training.

Real-World Data Science Applications

Key concepts • practical context • quick revision

- Healthcare: decision support, medical-image analysis and risk estimation with expert oversight.
- Retail: demand forecasts, product recommendations, inventory and customer segmentation.
- Finance: fraud detection, risk modeling and anomaly detection.
- Agriculture: crop disease recognition, yield estimation and irrigation planning.
- Transportation and smart cities: traffic prediction, route planning and energy optimization.
- Manufacturing: predictive maintenance from sensor signals and failure histories.

EXAMPLE / REMEMBER

Start from a measurable user need and define the cost of errors before modeling.

Deployment and MLOps

Key concepts • practical context • quick revision

- Deployment can be batch prediction, a web application, dashboard or prediction API.
- Monitor data drift, prediction quality, response time, failure rate and resource use.
- Version models and datasets; keep rollback options and document releases.
- Retraining should be triggered by evidence and validated before replacing a model.
- Protect endpoints with authentication, access controls and appropriate rate limits.
- Test input validation, reliability, security and behavior under changing workloads.

EXAMPLE / REMEMBER

A model's lifecycle continues after deployment: monitor, diagnose and improve.

Ethics, Privacy and Capstone Checklist

Key concepts • practical context • quick revision

- Collect only necessary data and apply consent, access controls, retention rules and security protections.
- Assess fairness across groups; look for biased labels, underrepresented samples and uneven error rates.
- Explain model limitations and uncertainty; provide human review for high-impact decisions.
- Capstone flow: define question → obtain data → clean → explore → baseline → train → evaluate → present.
- Deliverables: documented dataset, reproducible code, metrics, visualizations, limitations and recommendations.
- Success is measured by trustworthy impact and usefulness, not only model accuracy.

EXAMPLE / REMEMBER

Final question: would the result be safe, fair, explainable and useful in the real setting?